

Customer Churn Prediction System - Project Report

1. Title Page

- Project Title: Customer Churn Prediction System
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2. Acknowledgement

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Finally, I thank my classmates and family members for their encouragement and motivation during the development of this project.

3. Abstract

The Customer Churn Prediction System is designed to identify customers who are likely to leave a telecom service. In this project, the Telco churn dataset is analysed using Excel, MySQL, Python, Power BI, and machine learning techniques.

EDA was performed using charts like bar charts, heatmaps, pie charts, and box plots. SQL queries were used for churn analysis and business insights. Power BI dashboards were created for visualization.

The trained XGBoost machine learning model was integrated into a Streamlit web application where users can enter customer details and receive instant churn predictions. The system helps businesses improve customer retention and make data-driven decisions.

4. Introduction

Customer churn refers to customers leaving a company or discontinuing its services. In telecom industries, churn directly affects company revenue and growth.

By predicting churn in advance, companies can take preventive actions such as offering discounts, improving services, and increasing customer satisfaction. Machine learning helps in identifying patterns from historical customer data and provides accurate predictions.

5. Problem Statement

Companies face difficulty in identifying customers who are likely to leave their services. This project aims to predict customer churn using machine learning techniques to help organizations improve customer retention and reduce customer loss.

6. Objectives

- Predict whether a customer will churn or not
 - Improve customer retention strategies
 - Support business decisions using data
 - Perform customer behavior analysis
 - Provide a simple and user-friendly prediction system
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7. Data Collection and Dataset Description

Dataset Used:

Telco Customer Churn Dataset Dataset

File:

CSV_WA_Fn-UseC_-Telco-Customer-Churn.csv

Key Features:

- Tenure
- Monthly Charges
- Total Charges
- Contract Type
- Internet Service
- Payment Method
- Service Usage

Engineered Features:

- Tenure Group (New / Medium / Old)
 - Monthly Spending Category
 - Total Services Count
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8. Proposed System Architecture

Workflow:

Dataset → Excel Analysis → MySQL Analysis → Data Preprocessing → EDA → ML Model → Streamlit App → Prediction Output

Project Structure:

- Excel: Raw dataset analysis and pivot charts
- MySQL: SQL queries and churn analysis
- Python: EDA and preprocessing
- Machine Learning: XGBoost model
- Streamlit: Prediction web app
- Power BI: Dashboard visualization

9. Excel Analysis

Excel was used for:

- Data cleaning
- Pivot table analysis
- Contract-wise churn analysis
- Service-level analysis
- Chart creation

Charts Used:

- Pivot Tables
- Bar Charts
- Column Charts

Insights:

- Month-to-month customers had highest churn
 - Low service-level customers churned more
 - Fiber optic users showed higher churn rate
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10. MySQL Analysis

MySQL was used to perform customer churn analysis using SQL queries. SQL

Operations:

- Dataset analysis
- Churn count calculation
- Contract-wise churn analysis
- Internet service analysis
- Average monthly charges analysis
- Churn percentage calculation

Business Insights:

- Customers with short tenure churn more
 - High monthly charges increase churn risk
 - Long-term contracts reduce churn
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11. Python EDA and Machine Learning

EDA Charts Used:

- Count Plot
- Bar Chart
- Pie Chart
- Box Plot

- Heatmap
- Correlation Matrix

Libraries Used:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- XGBoost
- Joblib

Machine Learning Algorithm:

- XGBoost Classifier

Model Accuracy:

- Approximately 80%
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12. Power BI Dashboard

Power BI was used to create interactive dashboards and business visualizations. Dashboard

Visuals:

- Churn Distribution
- Contract-wise Churn
- Tenure Group Analysis
- Monthly Spending Analysis
- Payment Method Analysis
- KPI Cards

Dashboard Insights:

- Churn Rate: 26.54%
 - Month-to-month customers churned the most
 - Electronic check users showed higher churn behavior
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13. System Implementation

1. User opens Streamlit application
2. User enters:
 - Tenure , Monthly Charges , Total Services
3. Model processes input
4. Prediction is generated
5. Output displayed:
 - CHURN and NOT CHURN

14.Results

- Streamlit app successfully predicts churn
 - EDA identified important churn patterns
 - SQL analysis generated business insights
 - Power BI improved data visualization
 - Model achieved approximately 80% accuracy
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15.Benefits of the System

- Fast prediction
 - Helps improve customer retention
 - Supports decision-making
 - Interactive dashboard
 - Easy-to-use interface
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16. Limitations

- Limited input features in UI
 - Accuracy can be improved
- No live database connection
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17.Future Scope

- Improve model performance
 - Add more customer features
 - Deploy cloud-based application
 - Build mobile app version
 - Connect with live database
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18.Conclusion

This project successfully predicts customer churn using machine learning and provides a user-friendly Streamlit interface. The integration of Excel, MySQL, Python, Power BI, and XGBoost helped create a complete analytics and prediction system.

The project helps organizations identify potential customer loss and take preventive actions to improve customer retention.

19.References

- [Scikit-learn Documentation](#)
- [Streamlit Documentation](#)
- [Pandas Documentation](#)
- [NumPy Documentation](#)
- [Power BI Documentation](#)
- Telco Customer Churn Dataset (IBM/Kaggle)

20. Project Links

All project resources including Excel analysis, SQL scripts, Power BI dashboard, Python EDA and Streamlit application are available in the GitHub README file.